Research Statement

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Economic progress rests on the joint strength of markets and states—two systems that coordinate resources through very different mechanisms: prices and policies. Markets create prosperity when private incentives align with social welfare, and states sustain it when they can tax, regulate, and govern credibly. Yet in many economies—especially those marked by informality and institutional weakness—both falter in tandem. Markets are constrained by frictions that distort competition and investment, while states remain fiscally and politically fragile because much of economic activity lies beyond their reach. The result is a persistent equilibrium of low capacity and missed potential.

My research lies at this intersection. I ask a set of connected questions: *When can technology improve the state's ability to govern? When does it fail—and why? How do competitive forces within markets reinforce or undermine enforcement? And what role can citizens play in bridging the gap between state ambition and state capacity?* Guided by economic theory and tested through field and quasi-experimental methods, often in collaboration with governments and firms, I study how behavioral, technological, and organizational reforms can improve the functioning of markets and the capacity of the state.

Technology, Credibility, and the Limits of Digital State-Building

A foundational insight of my research is that information technologies—widely adopted as tools for "automated enforcement"—do not deliver durable compliance gains through deterrence unless the state can credibly act on the information they produce. In a working paper with Isabelle Cohen, I study the rollout of a real-time digital reporting system that transmits point-of-sale transaction data directly to the tax authority. Using administrative VAT panels matched to device-level adoption records, we document three findings that challenge the optimistic narrative around digital tax infrastructure.

First, adoption is sharply selected: firms that integrate the technology are already more tax-engaged before they adopt, complicating naïve estimates of the technology's impact. Second, conditional on adoption, the immediate rise in reported sales and tax payments fades within months, as firms update their beliefs about the state's ability to act on the data; the increase that does persist is smaller and comes from the convenience of a stable digital workflow and the preferential rate rather than from deterrence. Revelation without follow-up teaches firms that monitoring alone is hollow: what lasts is a lower cost of compliance, not a greater fear of detection.

Third, and most novel, the reform's own incentive architecture undermined the tax instrument it was meant to strengthen. Preferential tax rates were linked to restrictions on input-credit claims, which shifted firm reporting away from documenting input purchases. This weakened the self-enforcing chain of the VAT—the very mechanism that makes it, in theory, resistant to evasion. The lesson is not simply that "technology without teeth" fails; it is that poor design can actively erode the institutional logic of the tax system. Credibility and design—not technology alone—are central to digital state-building.

This paper anchors my broader agenda because it identifies the precise conditions under which information reforms fall short—motivating my subsequent projects that introduce enforcement, algorithmic tools, and citizen engagement as complements to monitoring.

Informality Traps: When Markets Undermine the State

If the e-POS study shows that technology alone cannot sustain compliance, a natural question follows: what happens when the state adds enforcement? The answer, it turns out, depends on the market. In ongoing work with Michael Best, Anders Jensen, and Adnan Khan, I develop and test the concept of an *informality trap*—a dynamic equilibrium in which weak enforcement and market competition reinforce each other. When evasion is widespread, compliant firms face a cost disadvantage; they are undercut by competitors who do not remit taxes. This makes compliance individually irrational even when collective formalization would benefit everyone. The state, in turn, collects too little revenue to invest in credible enforcement, and the trap persists.

This project is among the first to test experimentally whether such traps exist. Using a large-scale randomized enforcement intervention with tax authorities, we vary enforcement intensity across markets to examine whether coordinated pressure can break the cycle. The design connects insights from industrial organization—strategic complementarities, entry and exit dynamics—with core questions in public finance about the limits of state capacity. The intellectual contribution lies in showing that tax compliance is not merely an individual decision but a market-level outcome, shaped by the competitive environment firms inhabit.

Automation and Bureaucratic Judgment

A parallel strand of my research asks how the state's own internal organization adapts to new technology. In ongoing work with Adnan Khan, Ben Olken, and Mahvish Shaukat, I conduct a randomized evaluation that integrates *computer vision* into property tax administration. Using satellite and street-level imagery, a machine-learning algorithm predicts property assessments and flags properties for reassessment. The experiment introduces these algorithmic benchmarks to a random subset of inspectors, allowing us to study how automation reshapes bureaucratic judgment, effort allocation, and accountability.

The question this project asks is more fundamental than whether AI improves accuracy. It asks what happens to the human side of government when an algorithm can check—and challenge—a bureaucrat's work. Does the algorithm reduce discretion and improve consistency, or does it provoke strategic responses? Does it build citizen trust by making the assessment process more transparent, or does it erode the relational accountability that existed before? These questions sit at the intersection of public economics, organizational economics, and the rapidly growing literature on AI in government—and they have direct implications for how developing countries integrate emerging technologies into administrative systems.

Citizens as Monitors: Can Engagement from Below Strengthen Enforcement?

My e-POS study established that top-down monitoring without enforcement credibility is insufficient. The compliance traps project tests whether coordinated enforcement from above can shift the equilibrium. A third project completes this arc by asking whether pressure from *below*—from citizens themselves—can complement formal enforcement. In an ongoing randomized evaluation with Isabelle Cohen, I provide the first experimental evidence on *consumer receipt lotteries*, a mechanism used globally (in Brazil, China, Taiwan, Portugal, and several U.S. states) but never rigorously evaluated.

The design assigns business eligibility to participate in a lottery that rewards consumers for submitting formal sales invoices. Leveraging administrative VAT data, firm and consumer surveys, and mystery-shopping audits, we examine whether small incentives motivate consumers to demand receipts, whether this reduces buyer–seller collusion, and whether the intervention generates spillovers to non-participating firms. We also investigate changes in taxpayer–official interactions, rent-seeking behavior, tax morale, and perceptions of state capacity. The project directly tests whether citizen engagement can transform a monitoring technology—the same e-POS infrastructure I study in my job market paper—from a passive reporting tool into an active enforcement mechanism.

Social Identity and Inclusion in Markets

Markets do not operate in a social vacuum. Even as technology reduces transactional frictions, social norms and identity shape who participates, how they are treated, and what forms of exclusion persist. A key strand of my research examines these forces directly.

In a paper published in the *Journal of Economic Behavior and Organization*, I study racial discrimination in labor markets using a large-scale online experiment. I show that workers' productivity and willingness to cooperate depend not only on economic incentives but also on the social identity of their employers—revealing how social preferences generate inefficiencies even in competitive settings. Extending this inquiry to digital markets, my paper in the *Journal of Development Economics* examines gender dynamics in an online marketplace. Using experimental buyer profiles that signal gender, I find that while women face no discrimination in prices or product quality, they experience substantially more unsolicited contact from male sellers. Linguistic analysis reveals patterns of greater verbosity and flirtatious tone that, in conservative social contexts, impose reputational and social costs on women.

These findings matter for the broader agenda because they show that inclusion is not an automatic byproduct of market development or digitization. Understanding who is excluded from markets—and why—is essential for designing institutions that are both efficient and equitable.

Market Frictions, Climate Shocks, and Agricultural Networks

Just as institutional weaknesses limit how effectively states can govern, frictions and shocks within markets constrain economic performance. In ongoing work with Farah Said and Omar Gondal, I study how *information asymmetries* and *entrenched trading relationships* restrict firms' ability to respond to prices and realize potential gains from trade. Even when arbitrage opportunities exist, market participants often fail to exploit them because of relational dependencies and informational barriers.

A complementary project uses high-frequency trade data and remote-sensing imagery to examine how the catastrophic 2022 floods in Pakistan disrupted agricultural supply networks. We find sharp reductions in trade flows between rural producers and urban consumers, with significant heterogeneity across crop types and district exposure. Notably, anticipatory effects—farmers increasing supply before the flood—appear only in historically high-risk regions, suggesting that prior experience with shocks shapes adaptive behavior. The results underscore that the economic impact of climate events depends not only on physical damage but also on the flexibility and structure of the market institutions that mediate production and exchange.

Future Directions

Across these projects, my aim is to build a cumulative account of how markets and states co-evolve: when reductions in frictions on the exchange side translate into durable gains only if paired with credible, low-cost enforcement, and when stronger enforcement is effective only if it preserves competition and inclusion. The arc of my current work—from the limits of technology, to the market dynamics that entrench informality, to the role of citizens in strengthening enforcement—reflects a unified empirical program on the institutional foundations of state effectiveness.

Looking ahead, I am extending this agenda in several directions. One project will investigate the determinants of whistleblowing—who chooses to report misconduct and how citizen reporting can be institutionalized as a credible accountability mechanism. Another examines customer-driven discrimination in digital marketplaces, studying how demand-side behavior constrains women's participation and what platform design features might mitigate these barriers. A third evaluates the implementation of a national *track-and-trace* system that uses digital monitoring to improve supply-chain compliance—extending my work on the VAT chain to a broader set of goods and enforcement architectures. Together, these projects advance a broader inquiry into how technological and institutional innovations can strengthen the social foundations of effective and equitable governance.

More broadly, my ambition over the next decade is to move from documenting when reforms succeed or fail toward developing a predictive framework for *institutional design*—one that specifies the conditions under which different combinations of technology, enforcement, and citizen engagement are likely to deliver durable improvements in state capacity and market performance. The evidence base I am building—spanning field experiments, administrative panels, and frontier measurement tools—is designed to support exactly that kind of cumulative knowledge.

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